

Data Frames in 3.0-Preprocessing

Data Frame	Description and evolution
dfs	Contains the solar array data. Columns were renamed to simpler names, and some irrelevant columns were dropped.
dfp	Contains the US population data for all counties. The data was filtered to include California only, and all columns were dropped except for county and population density.
dgeo	Contains the geometric information of the counties in dfs (latitude and longitude).
dfs_temp	Left merge of dfs and dgeo
df	Left merge of dfs_temp and dfp. Final data frame before encoding and split.
df_enc	Copy of df, used to encode the categorical variables. Variables representing percentiles were ordinally encoded (0-3). The install type was dummy encoded. Urban/Rural and the target variable STI were binary encoded ("Urban":0,"Rural":1, "Within":1,"Outside":0). The variables before encoding and the county variable was dropped.
X, y, X_train, X_test, y_train, y_test	The usual feature/target and train/test split. Longitude and latitude columns are dropped from X_train and X_test due to low correlation with target variable.
X_train_os, y_train_os	Train data after being resampled with an oversampling technique.
X_train_us, y_train_us	Train data after being resampled with an undersampling technique.
X_train_os_robust, X_test_robust	X data after being resampled with an oversampling technique AND rescaled with robust method.
X_train_us_robust, X_test_robust	X data after being resampled with an undersampling technique AND rescaled with robust method.
X_train_os_minmax, X_test_minmax	X data after being resampled with an oversampling technique AND rescaled with minmax method.